

SESSION 1

RAD-447 Radionuclide Imaging

Foundations of Radionuclide Imaging

Radionuclides, radiopharmaceuticals, radioactive decay and half-life

Course	RAD-447 Radionuclide Imaging / Nuclear Medicine
Date	2026
Lecturer	A. A. Zubair
Suggested study time	180–240 minutes

Learning Objectives

By the end of this session, the student should be able to:

1. define radionuclide imaging and explain why it is primarily a functional imaging technique;
2. distinguish diagnostic from therapeutic nuclear medicine;
3. define a radionuclide and a radiopharmaceutical and explain the role of each component;
4. describe the principal types of radioactive decay relevant to nuclear medicine;
5. explain physical half-life and calculate the remaining activity after successive half-lives; and
6. explain why emitted photons must be able to leave the patient if they are to be detected externally.

Key Concept

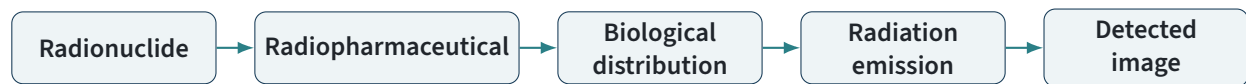
Radionuclide imaging images physiology more directly than anatomy. A radiopharmaceutical participates in, or follows, a biological process. The detector then maps where the radioactive component is located. The resulting image therefore reflects tracer distribution, which is related to functions such as blood flow, metabolism, receptor binding, organ uptake or excretion.

1 Session Orientation

This session provides the foundation for the rest of the course. Before learning how a gamma camera, SPECT system or PET scanner forms an image, the student must first understand three linked questions:

1. **What is administered?** — a radiopharmaceutical containing a radionuclide;

2. **Why does it localize in a particular tissue?** — because of its biological and chemical behaviour; and
3. **How can its distribution be imaged?** — because radioactive decay produces radiation that can be detected outside the patient.



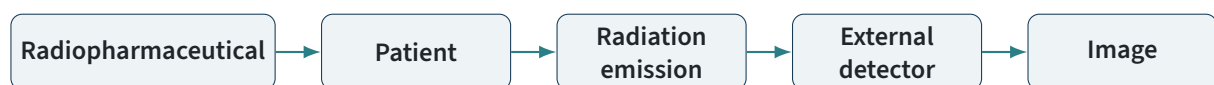
Why this foundation matters

Every later nuclear-medicine examination can be understood by asking: **Which tracer was used? What physiological process controls its biodistribution? What radiation is emitted? How is that radiation detected?** These questions connect tracer biology to image formation and clinical interpretation.

2 What Is Radionuclide Imaging?

Radionuclide imaging is a branch of medical imaging in which a radioactive substance is administered to the patient and the radiation emitted from within the body is detected to form diagnostic images.

This direction of radiation is fundamentally different from conventional X-ray imaging. In radiography and CT, radiation originates from an external X-ray tube and passes through the patient. In radionuclide imaging, the radioactive source is placed *inside* the patient as part of a radiopharmaceutical, and the detector records radiation emerging from the body.



The complete imaging process can be summarized as:

Radionuclide → Radiopharmaceutical → Patient → Radiation emission → Detector → Image → Interpretation

2.1 Function versus structure

CT and MRI are particularly strong at showing structural relationships and anatomical detail. Radionuclide imaging is especially valuable when the clinical question concerns **how tissue is functioning** rather than only what it looks like.

Examples include bone turnover, renal function and drainage, thyroid activity, myocardial perfusion, hepatobiliary excretion, pulmonary ventilation/perfusion and glucose metabolism.

High-yield distinction

A useful conceptual contrast is:

CT/MRI: What does the organ look like?

Radionuclide imaging: How is the organ functioning?

This is a simplification, not an absolute rule. CT and MRI can provide functional information, but radionuclide imaging is built around tracer physiology and molecular distribution.

2.2 Why functional imaging can reveal disease early

Many diseases disturb physiology before they produce a large structural change. Altered perfusion, metabolism, receptor expression, bone turnover or organ clearance may therefore become abnormal before a lesion is conspicuous on purely anatomical imaging. This is one reason radionuclide studies can provide clinically useful information even when structural change is subtle.

This does not mean radionuclide imaging is always more sensitive or more specific than CT or MRI. The value of each examination depends on the clinical question, the tracer used and the biological process being measured. Modern hybrid systems often combine both approaches so that functional information can be localized anatomically.

3 Nuclear Medicine: Diagnosis and Therapy

Nuclear medicine is the medical specialty that uses radioactive materials for diagnosis and, in selected circumstances, treatment.

3.1 Diagnostic nuclear medicine

Diagnostic examinations use relatively small administered activities to obtain information about tracer distribution and physiology. Common diagnostic examples include:

- bone scintigraphy;
- thyroid imaging;
- renal scintigraphy;
- myocardial perfusion imaging; and
- PET/CT.

3.2 Therapeutic nuclear medicine

Therapeutic nuclear medicine deliberately uses radiation to deliver a clinically useful absorbed dose to targeted tissues. Examples include iodine-131, lutetium-177, yttrium-90 and radium-223 therapies.

Clinical / Practical Link

The same broad principle underlies both diagnosis and therapy: a radioactive substance is delivered to a biological target. Diagnostic nuclear medicine emphasizes **detectability and imaging**; therapeutic nuclear medicine emphasizes **energy deposition in target tissue**.

4 Radionuclides

A **radionuclide** is an unstable atomic nucleus that undergoes radioactive decay. During decay, energy is released as particles and/or electromagnetic radiation.

4.1 Isotope, radionuclide and activity

Atoms of the same element have the same number of protons but may contain different numbers of neutrons; these forms are called **isotopes**. An isotope that is unstable and radioactive is a **radionuclide**. Thus, the term radionuclide emphasizes radioactive instability rather than merely atomic identity.

The amount of radioactivity present is described by **activity**, which means the number of nuclear transformations occurring per unit time. The SI unit is the **becquerel (Bq)**, where 1 Bq represents one nuclear transformation per second. In clinical nuclear medicine, activities are commonly expressed in kilobecquerels (kBq), megabecquerels (MBq) or gigabecquerels (GBq).

Activity is not the same as radiation dose

Activity describes how rapidly nuclei are transforming. It does **not**, by itself, state how much energy is absorbed by tissue. Patient dose also depends on the type and energy of radiation, tracer biodistribution, residence time and the organs exposed.

Several radionuclides appear repeatedly in nuclear medicine:

Radionuclide	Approx. half-life	Common nuclear-medicine application
Technetium-99m	6 h	Conventional gamma-camera and SPECT imaging
Fluorine-18	110 min	PET imaging
Iodine-123	13 h	Thyroid imaging
Gallium-67	78 h	Infection/tumour imaging
Thallium-201	73 h	Myocardial imaging
Carbon-11	20 min	PET research/neurology
Nitrogen-13	10 min	PET
Oxygen-15	2 min	PET research

4.2 Why technetium-99m is so important

Technetium-99m (Tc-99m) is the most widely used radionuclide in conventional gamma-camera and SPECT imaging. Its usefulness reflects a favorable combination of properties:

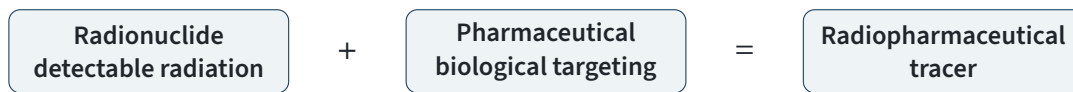
- a relatively short physical half-life;
- gamma emission suitable for external detection;
- relatively low patient radiation burden compared with many alternatives;
- practical availability through a radionuclide generator; and
- the ability to be incorporated into many different radiopharmaceuticals.

Common Error

Do not use **radionuclide** and **radiopharmaceutical** as synonyms. Tc-99m is a radionuclide. Tc-99m MDP is a radiopharmaceutical. The distinction matters because the radioactive atom provides the detectable radiation, while the complete radiopharmaceutical determines the biological behavior of the tracer.

5 Radiopharmaceuticals

A **radiopharmaceutical** combines a radioactive component with a pharmaceutical or biologically active component.



The pharmaceutical component influences where the tracer travels, how it is taken up, and how it is cleared. The radionuclide provides the radiation that allows the distribution of that tracer to be measured externally.

Radiopharmaceutical localization can occur through several physiological mechanisms. Some tracers follow **blood flow**; some participate in or resemble a **metabolic substrate**; some bind to a **receptor or transporter**; and others are handled according to a specific organ's **uptake, secretion or excretion pathway**. The image therefore makes sense only when the expected tracer behaviour is understood.

5.1 Example 1: Tc-99m MDP

Tc-99m is the radioactive component. MDP is the bone-seeking pharmaceutical component. The compound localizes predominantly in bone and is used for bone scintigraphy. Its distribution reflects physiological factors such as regional blood flow and osteoblastic activity, which will be developed in the clinical-applications session.

5.2 Example 2: F-18 FDG

F-18 is a positron-emitting radionuclide. FDG is a glucose analogue. Its distribution is related to tissue glucose metabolism, which makes FDG-PET useful in several areas, especially oncology, neurology and cardiology.

Think through the tracer

Suppose two patients receive the same radionuclide but attached to different pharmaceutical compounds. Would the images necessarily show the same organs? **No.** The radionuclide determines the emitted radiation, but the pharmaceutical component strongly influences where the tracer localizes. Nuclear medicine therefore depends on both nuclear physics and biological targeting.

6 Biodistribution: Why Tracers Go Where They Go

Biodistribution describes how a radiopharmaceutical distributes, localizes and clears within the body. It is a central concept because tracer distribution is what gives a radionuclide image its physiological meaning.

Tracer distribution can depend on:

- blood flow;
- cellular metabolism;
- receptor binding;
- organ function; and
- excretion pathways.

Example tracer	Dominant biological idea	What the image can reflect
Tc-99m MDP	Bone turnover and regional delivery	Areas of relatively increased or decreased skeletal tracer uptake
F-18 FDG	Glucose use / metabolic trapping	Relative tissue glucose metabolism
Renal radiopharmaceutical	Renal perfusion, uptake and/or excretion	Renal function and urinary drainage
Thyroid tracer	Thyroid trapping / handling	Regional thyroid functional activity

Core biological principle

A nuclear medicine image is not merely a map of radioactivity. It is a map of **radioactivity shaped by biology**. To interpret an image properly, the observer must know the expected biodistribution of the radiopharmaceutical and understand the physiological process that controls uptake and clearance.

7 Radioactive Decay

Radioactive nuclei are unstable. They transform toward more stable states through **radioactive decay**. The type of emission produced determines whether a radionuclide is more suitable for imaging, therapy, or both.

Decay / emission	What is emitted?	Main relevance in nuclear medicine
Alpha decay	Alpha particle	Mainly therapeutic rather than routine diagnostic imaging
Beta-minus decay	Electron (β^-)	Mainly therapeutic nuclear medicine
Positron emission	Positron (β^+)	Fundamental to PET
Gamma emission	Gamma photon	Fundamental to gamma-camera and SPECT imaging

The practical distinction is related to how far the emitted radiation travels and where its energy is deposited. Alpha and beta particles have relatively short ranges in tissue and are therefore particularly useful when the aim is to deposit energy locally for therapy. Gamma photons can travel farther and may escape the body, making external detection possible. Positrons travel only a short distance before annihilating with electrons; PET detects the two annihilation photons produced by that interaction rather than the positron itself.

7.1 Why gamma photons are useful for imaging

Gamma photons are electromagnetic radiation emitted from the nucleus. Because photons can travel out of the patient and reach an external detector, they can be used to infer where the radiopharmaceutical is located.

Tc-99m → gamma photon → gamma camera

The detector does not directly “see” the organ. It detects emitted radiation and uses the location and characteristics of those detected events to build an image of tracer distribution.

Not every emitted photon reaches the detector. Some photons are absorbed or scattered within the patient before leaving the body. These effects reduce detected counts and can alter measured spatial distribution. Attenuation and scatter therefore become important image-quality and correction topics later in the series.

High-Yield

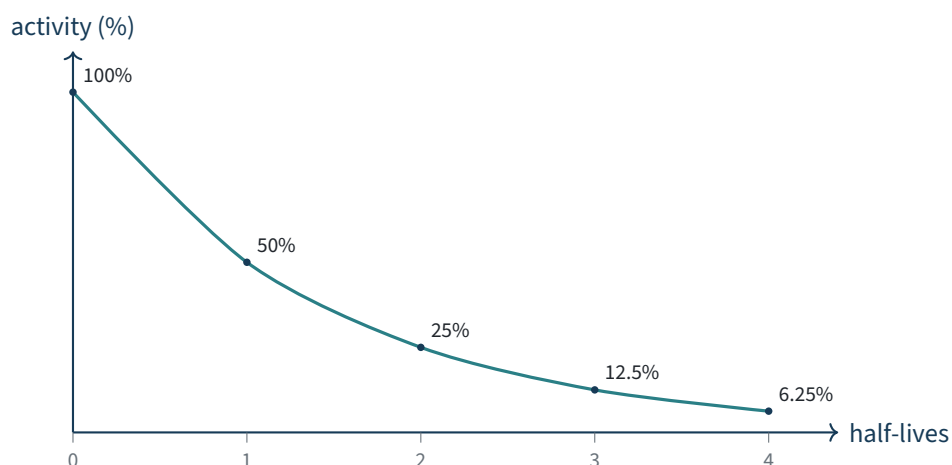
SPECT and conventional gamma-camera imaging depend on detecting single emitted gamma photons. **PET** depends on positron emission followed by annihilation, which produces two 511-keV photons. PET physics will be treated separately in Session 3.

8 Physical Half-Life

The **physical half-life** is the time required for half of the radioactive nuclei in a sample to decay.

For Tc-99m, the physical half-life is approximately 6 hours. Therefore:

Time after reference point	Activity remaining
0 h	100%
6 h	50%
12 h	25%
18 h	12.5%
24 h	6.25%



8.1 Why half-life matters clinically

A diagnostic radionuclide must remain radioactive long enough for preparation, administration, uptake and imaging. At the same time, unnecessarily long persistence would increase radiation exposure without adding diagnostic benefit.

Physical half-life describes only nuclear decay. A radiopharmaceutical may also be removed from an organ or from the body through biological processes. Later in the course this distinction will be developed into **physical half-life**, **biological half-life** and **effective half-life**. For now, remember that physical decay and biological clearance are separate mechanisms.

Successive half-lives

A sample contains 800 MBq at a reference time. Ignore biological elimination. After one physical half-life, 400 MBq remains. After two half-lives, 200 MBq remains. After three half-lives, 100 MBq remains. The amount halves during each equal half-life interval; it does not decrease by a fixed number of MBq each time.

Clinical / Practical Link

Half-life affects practical workflow. A very short-lived radionuclide can demand rapid synthesis, transport and imaging. A longer-lived radionuclide is easier to distribute but remains radioactive for longer. Nuclear medicine therefore balances physical suitability, biological behavior, logistics and radiation protection.

Exponential decay relationship

The exponential decay relationship can be written as

$$A(t) = A_0 \left(\frac{1}{2} \right)^{t/T_{1/2}},$$

where A_0 is the initial activity and $T_{1/2}$ is the physical half-life. This equation is useful when the elapsed time is not an exact whole number of half-lives.

For example, after 9 hours a Tc-99m sample has passed through 1.5 half-lives. The remaining physical activity is therefore about 35% of the starting value, not 25% or 50%. The equation is especially useful for such intermediate times.

9 From Emission to Detection: A Bridge to the Gamma Camera

The gamma camera is the principal detector system for conventional planar nuclear medicine and SPECT. Its detailed operation belongs in Session 2, but the basic logic should already be clear:



The gamma camera must answer two practical questions about each accepted event: **where did the photon interact?** and **was its energy consistent with the radionuclide being imaged?** Session 2 develops how the collimator, NaI(Tl) crystal, photomultiplier tubes and pulse-height analysis accomplish this.

Key Takeaways

- Radionuclide imaging detects radiation emitted from a radiopharmaceutical located inside the patient.
- Its major strength is the depiction of physiological, metabolic and molecular processes.
- A radionuclide is the unstable radioactive atom; a radiopharmaceutical is the complete tracer used clinically.
- The pharmaceutical component largely determines biological distribution, while the radionuclide provides detectable radiation.
- Gamma emission is central to conventional gamma-camera/SPECT imaging, while positron emission is central to PET.
- Physical half-life describes how quickly radioactivity falls because of nuclear decay.
- Tc-99m is a major SPECT radionuclide because its physical and practical properties are well suited to diagnostic imaging.

Key Terms

Term	Meaning
Radionuclide imaging	Imaging based on radiation emitted from radioactive material administered to the patient.
Nuclear medicine	Medical specialty using radioactive substances for diagnosis and selected therapies.
Radionuclide	Unstable atomic nucleus that undergoes radioactive decay.
Radiopharmaceutical	Radioactive tracer combining a radionuclide with a biologically directed pharmaceutical component.
Biodistribution	Pattern of tracer distribution, uptake and clearance in the body.
Gamma photon	Electromagnetic radiation emitted from the nucleus and detectable outside the patient.
Positron	Positively charged antiparticle of the electron; central to PET radionuclides.
Physical half-life	Time required for half of the radioactive nuclei in a sample to decay.

Self-Assessment Questions

1. Define radionuclide imaging and explain why it is commonly described as functional imaging.
2. Distinguish a radionuclide from a radiopharmaceutical using Tc-99m MDP as an example.
3. State four factors that can influence radiopharmaceutical biodistribution.
4. Compare the relevance of alpha, beta-minus, positron and gamma emissions in nuclear medicine.
5. Tc-99m has a physical half-life of about 6 hours. What fraction of the original activity remains after 18 hours, and why is this clinically relevant?
6. State five reasons Tc-99m is widely used in SPECT imaging.

Series Navigation

1	Foundations of Radionuclide Imaging — this session
2	Gamma Camera, Planar Imaging & SPECT
3	PET, Attenuation Correction & Hybrid Imaging
4	Clinical Radionuclide Imaging Applications
5	Image Quality, Radiation Protection, Workflow, QC, Artifacts & Future Directions

